

Muhammad Omair Tahir

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EDUCATION

Comsats University Islamabad <i>B.Sc Computer Science (CGPA: 3.48)</i>	Islamabad, Pakistan <i>Feb. 2023 – Jan. 2027</i>
Beaconhouse College Program <i>A-Levels in Natural Sciences</i>	Islamabad, Pakistan <i>Aug. 2020 – May 2022</i>

Professional Summary

AI Engineer with hands-on experience in building deep learning models, optimizing CNN and Transformer architectures, and working with large datasets. Skilled in Python, PyTorch, and ML workflows, with a strong foundation in algorithms, data structures, and applied AI research. Passionate about developing production-ready AI solutions and transforming business processes through intelligent automation.

EXPERIENCE

AI Intern <i>Medical Imaging and Diagnostic Lab, NCAI</i>	July 2025 – Sept. 2025 <i>Onsite</i>
- Built and optimized a two-stage breast cancer classification model using VinDr-Mammography dataset. - Evaluated multiple CNN and Transformer architectures and finalized DenseNet121 with full preprocessing, achieving high model accuracy. - Conducted performance analysis and data augmentation to improve model robustness.	

PROJECTS

Mammography Classification <i>Computer Vision, Machine Learning & Deep Learning</i>	Sept. 2025
- Developed a two-stage breast cancer classification pipeline with DenseNet121. - Implemented preprocessing, augmentation, and evaluation for robust model performance.	
Multi-Document Research Assistant <i>NLP, Retrieval-Augmented Generation (RAG), Transformers</i>	Jan. 2025
- Integrated RoBERTa-based QA model for extractive answer generation with confidence-based ranking. - Developed Flask API supporting multi-document indexing and natural language querying.	

Professional Certifications

Unsupervised Learning in Python Skills: Cluster Analysis, Feature Extraction, SciPy
Supervised Learning with scikit-learn Skills: scikit-learn, pandas, NumPy
Machine Learning Specialization Skills: Supervised Learning, Unsupervised Learning, Advanced Learning Algorithms

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL
Frameworks/Libraries: PyTorch, Flask, React, Next.js, NumPy, Matplotlib, difflib
AI/ML: CNNs, Transformers, Model Training, Evaluation, Data Augmentation, Retrieval- Augmented Generation (RAG), Transformers, Natural Language Processing (NLP), Reinforcement Learning (basic), Feature Engineering
Databases: MongoDB
Tools: Git, VS Code, PyCharm, Visual Studio, IntelliJ, Excel, PowerPoint, Outlook

Extracurricular & Volunteer Experience

GDG on Campus CUI – Programming Co-Lead	Sept. 2024 – Feb. 2025
Led programming workshops and coding sessions, mentored students, and coordinated hackathon activities.	
GDG on Campus CUI – Social Media Lead	Feb. 2024 – Aug. 2024
Managed social media campaigns reaching 150+ students, tracked engagement KPIs, and optimized content strategy based on performance metrics	